

9 Monte Carlo Methods

9.1 Accept-Reject Method

There are many distributions where the inverse transform method cannot be applied. For such cases, we shall now discuss some indirect methods. The ‘Accept-Reject algorithm’ is an excellent indirect method when known transformation methods are not available. The general idea is to generate a candidate random variable from some other distribution and accept it if it passes a test. This class of ‘Accept-Reject algorithm’ is remarkably robust. It allows us to simulate from virtually any distribution.

Accept-Reject Methods:

- Suppose f is the **target density**, from which we want to simulate; but we don’t know how.
- We take g as the **candidate density**. It should be chosen in such a way that it is simple to simulate random variables from g .
- We need to put another constraint on g . In mathematical terms, the probability measure associated with f is absolutely continuous with respect to the probability measure associated with g . In layman terms, this translates to $f(x) > 0$ if $g(x) = 0$.
- The algorithm to simulate a random observation $X \sim f$ goes as follows:
 1. Generate $Y \sim g$ and $U \sim \text{unif}(0, 1)$ independently.
 2. If $U \leq \frac{1}{M} \frac{f(Y)}{g(Y)}$ set $X = Y$, that is accept the candidate random variable Y . Else, discard Y and U and start again.
- We do lose a lot of candidate random variables in the process. If $M = \sup_x \frac{f(x)}{g(x)}$, then the probability of accepting a particular candidate variable is $\frac{1}{M}$ and the expected waiting time between successive acceptances is M . Thus to design an efficient algorithm, we need to choose g so that M is small.

Why does this method work?

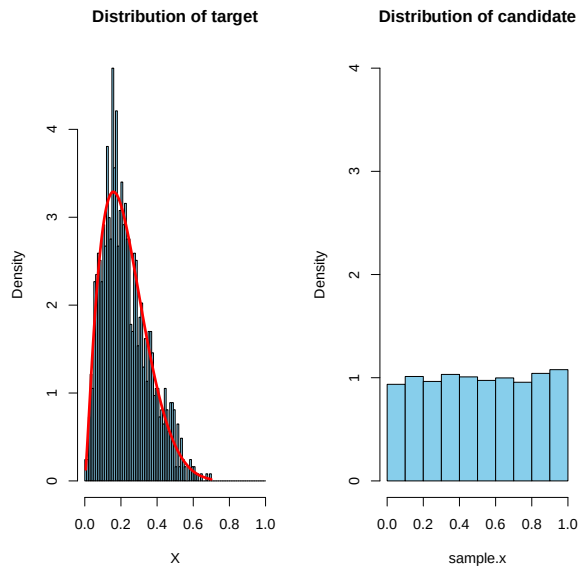
A straightforward probability calculation shows

$$P(Y \leq x | \text{Accept}) = P\left(Y \leq x | U \leq \frac{f(Y)}{Mg(Y)}\right) = P(X \leq x),$$

where $Y \sim g$ and $X \sim f$. Simulating from g , the output of this algorithm is exactly distributed from f .

The **Accept-Reject method** is applicable in any dimension. As long as g is a density over the same space as f . Only need to know f/g upto a constant and require an upper bound on M .

Accept-Rejection Algorithm for Beta Distribution Generate $X \sim \text{Beta}(a=2.3, b=7.9)$ using Accept-Rejection algorithm



1. **Candidate distribution** $Unif(0, 1)$
2. **Target distribution:** $Beta(2.3, 7.9)$

9.2 Importance Sampling

Importance sampling is used to estimate probabilities and integrals. It is based on an alternative formulation of the Laws of Large Number (LLN). Let us assume

1. f is the target density,
2. g is the candidate density.

Objective is to estimate $\mathbb{E}_f(h(Y))$, such that

$$\begin{aligned}
 E_f(h(Y)) &= \int h(y)f(y)dy \\
 &= \int h(y)\frac{f(y)}{g(y)}g(y)dy \\
 &= E_g\left[\frac{h(X)f(X)}{g(X)}\right]
 \end{aligned}$$

So in '**Importance Sampling**' all you do is first you generate samples from candidate distribution g . Suppose $(\theta^1, \theta^2, \dots, \theta^N)$ are the random samples

generated from g . You estimates

$$\bar{h} = \frac{1}{N} \sum_{i=1}^N \frac{f(\theta^i)}{g(\theta^i)} h(\theta^i)$$

and by virtue of LLN we have

$$\bar{h} = \frac{1}{N} \sum_{i=1}^N \frac{f(\theta^i)}{g(\theta^i)} h(\theta^i) \longrightarrow E_g \left[\frac{h(\theta)f(\theta)}{g(\theta)} \right] = E_f(h(\theta)).$$

The logic underlying the ‘importance sampling’ lies in a simple rearrangement of terms in the target integral and multiplying by 1:

$$\int h(\theta)f(\theta)d\theta = \int h(\theta)\frac{f(\theta)}{g(\theta)}g(\theta)d\theta = \int h(\theta)\omega(\theta)g(\theta)d\theta$$

here $g()$ is another density whose support is same as $f()$ and we know how to simulate from $g()$.

The $\omega()$ is called ‘importance function’. Put more mass where h is large. So $w = f/g$ is smaller where h is larger.

Objectives of Importance sampling

1. Estimate probabilities of rare events
2. Reduce the variance of an integral approximation.
3. Evaluating expectation when you cannot generate from the density f
4. Draw inference or generate sample when the target density is unnormalized

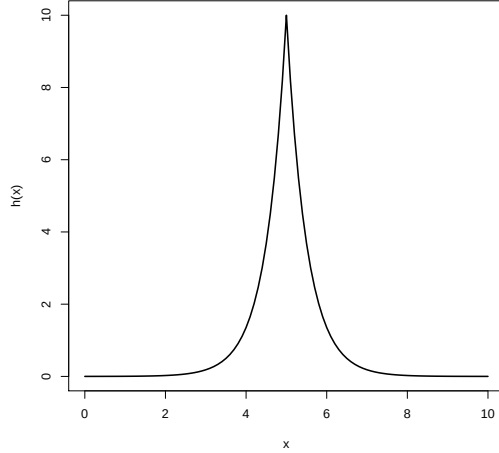
Ex: Estimation of Rare Event Probability with the Importance Sampling

$Y \sim N(0, 1)$. This is f , the target density. We want to estimate $P(Y < -5)$. The probability is expressed in log-scales, i.e., $\ln(P(Y < -5))$.

1. Simulating $Z^{(i)}$ from $N(0, 1)$ only produces a hit once in about 3 million iterations !
2. This is a very **rare event** for standard normal distribution
3. Estimates from Importance sampling is done by simulating from g which is $N(-5, 1)$.

Reduce the variance of an integral approximation The simplest Monte Carlo method to approximate an integral $\theta = \int h(y)f(y)dy$, where f is a pdf is to generate random variables $Y_1, \dots, Y_n$ from f and use the estimator $\hat{\theta} = \sum_{i=1}^n h(Y_i)$. This is consistent and unbiased.

$$Var(\hat{\theta}) = \int h^2(y)f(y)dy - \theta^2$$



An importance sampling estimator with candidate density g is $\tilde{\theta} = \sum_{i=1}^n h(X_i) \frac{f(X_i)}{g(X_i)}$ where $X_1, \dots, X_n$ are iid g . This is also consistent and unbiased.

$$\text{Var}(\tilde{\theta}) = \int h^2(y) \frac{f(y)}{g(y)} f(y) dy - \theta^2$$

Thus $\text{Var}(\tilde{\theta}) - \text{Var}(\hat{\theta}) = \int h^2(y) (\frac{f(y)}{g(y)} - 1) f(y) dy$ is negative if $f(y) < g(y)$ that is $\frac{f(y)}{g(y)} - 1 < 0$ for large values of h . Note that both f and g integrate to 1, so we cannot have $\frac{f(y)}{g(y)} - 1 < 0$ for all y . So g should put larger mass where h is larger.

Ex: Importance Sampling to improve integral approximation

Consider the function $h(x) = 10 \exp(-2|x - 5|)$. Suppose that we want to calculate $E(h(X))$, where $X \sim \text{Uniform}(0, 10)$. That is we want to calculate

$$\int_0^{10} \exp(-2|x - 5|) dx.$$

The true value for this integral is about 1. The simple way to do this is to generate x_i from the $\text{Uniform}(0, 10)$ and look at the sample mean of $h(x_i)$. Notice this is equivalent to importance sampling with importance function $\omega(x) = f(x)$, where $f(x) = \frac{1}{10} I(0 < x < 10)$. Note that $h(x) = 10 \exp(-2|x - 5|)$ where $x \sim \text{Uniform}(0, 10)$.

Note that $h(x) = 10 \exp(-2|x - 5|)$ where $x \sim \text{uniform}(0, 10)$. The function $h(\cdot)$ in this case is peaked at 5, and decays quickly elsewhere, therefore, under the uniform distribution, many of the points are contributing very little to this expectation. Something more like a gaussian function ce^{-x^2} with a peak at 5 and small variance, say, 1, would provide greater precision. We can re-write the

integral as

$$\int_0^{10} 10 \exp(-2|x-5|) \frac{1/10}{\frac{1}{\sqrt{2\pi}} e^{-(x-5)^2/2}} \frac{1}{\sqrt{2\pi}} e^{-(x-5)^2/2} dx.$$

We want to estimate $\mathbb{E}[h(X)\omega(X)]$ where $X \sim N(5, 1)$. So in this case $f(x) \sim \text{Uniform}(0, 10)$, $g(x) \sim N(5, 1)$. The weight function is

$$\omega(x) = \frac{\sqrt{2\pi} e^{(x-5)^2/2}}{10}.$$

The example shows that the choice of correct weight function is important to reduce the Monte Carlo error.

Ex: Estimation of expectation when the target density is known upto a constant Assume that the target density f is only known upto proportionality, i.e. $f(x) = f_u(x)/c_f$ where f_u is a known unnormalized density and c_f is an unknown normalizing constant. Moreover, assume that the candidate density is $g(x) = g_u(x)/c_g$ where g_u is a known unnormalized density and c_g is a (possibly unknown) normalizing constant. Then the estimator of $\theta = \int h(x)f(x)dx$ is

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n h(Y_i) f(Y_i) / g(Y_i) = \frac{1}{n} \sum_{i=1}^n h(Y_i) \frac{f_u(Y_i)}{g_u(Y_i)} \frac{c_f}{c_g}$$

where $Y_1, \dots, Y_n$ are generated from g . depends on the ratio c_g/c_f of unknown normalizing constants. Note that

$$\frac{c_f}{c_g} = \frac{\int f_u(x) dx}{c_g} = \int \frac{f_u(x)}{g_u(x)} \frac{g_u(x)}{c_g} dx = \int \frac{f_u(x)}{g_u(x)} g(x) dx = E \frac{f_u(Y)}{g_u(Y)}$$

where Y follows the candidate density g . Thus it is possible to estimate c_f/c_g , using the sample $Y_1, \dots, Y_n$.

$$\widehat{\frac{c_f}{c_g}} = \frac{1}{n} \sum_{i=1}^n \frac{f_u(Y_i)}{g_u(Y_i)}$$

Finally we obtain the following consistent estimator of θ :

$$\hat{\theta} = \frac{\sum_{i=1}^n h(Y_i) \frac{f_u(Y_i)}{g_u(Y_i)}}{\sum_{i=1}^n \frac{f_u(Y_i)}{g_u(Y_i)}}$$